

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)
FR-1	Functional	The system shall allow the user to select a file that needs to be compressed.	High	When a valid file is selected, the system accepts it as the input for compression.	Required to initiate compression.
FR-2	Functional	The system shall read the input file and calculate the frequency of each character or byte.	High	For a selected file, the system produces frequency information for the characters or bytes present in the file.	Required for Huffman tree construction.
FR-3	Functional	The system shall construct a Huffman tree based on the calculated character or byte frequencies.	High	A Huffman tree is generated from the frequency data for the selected input file.	Core Huffman coding operation.
FR-4	Functional	The system shall generate variable-length binary Huffman codes for each character or byte using the constructed tree.	High	Each character or byte represented in the tree is assigned a corresponding Huffman code.	Required for encoding the input.
FR-5	Functional	The system shall encode the original file using the generated Huffman codes and create a compressed file.	High	After compression, a compressed output file is created successfully.	Primary purpose of the system.

FR-6	Functional	The system shall store the compressed data together with the information required to reconstruct the Huffman tree during decompression.	High	A valid compressed file contains sufficient information for the system to reconstruct the required Huffman tree.	Required for successful decompression.
FR-7	Functional	The system shall decode a valid compressed file using the Huffman tree and recreate the original file.	High	When a valid compressed file is decompressed, the original file is recreated.	Provides the reverse compression operation.
FR-8	Functional	The system shall display the original file size, compressed file size, compression ratio, and percentage of space saved.	Medium	After compression, all four compression results are displayed to the user.	Allows the user to evaluate compression effectiveness.
FR-9	Functional	The system shall allow the user to view the generated Huffman codes and visualize the Huffman tree.	Low	The system displays the generated codes and a representation of the Huffman tree for the selected file.	Helps the user understand the compression process.
FR-10	Functional	The system shall detect and handle invalid files, corrupted compressed files, and unsuccessful compression or decompression operations.	High	When an invalid or corrupted input is encountered, the system reports the failure instead of producing an invalid result.	Prevents silent failures and protects data integrity.

NFR-1	Nonfunc.	Compression and decompression should be performed within a reasonable amount of time.	High	Compression and decompression complete within an acceptable time for supported file sizes.	Ensures efficient operation.
NFR-2	Nonfunc.	The compressed output should reduce storage space compared with the original file, particularly for files containing repeated data.	High	For files that are suitable for Huffman compression, the compressed output is smaller than the original where compression overhead permits.	Provides the intended storage-saving benefit.
NFR-3	Nonfunc.	The system should consistently decompress a valid compressed file back to the original data without loss.	High	Repeated decompression tests on valid compressed files reproduce the original data without loss.	Ensures dependable lossless recovery.
NFR-4	Nonfunc.	Huffman codes and the resulting compressed data should be generated correctly without altering the original information.	High	The generated codes and compressed output correctly represent the original input and support accurate decompression.	Ensures correctness of the algorithm.
NFR-5	Nonfunc.	The interface should be simple enough for a user to select, compress, and decompress files easily.	Medium	A user can complete file selection, compression, and decompression through the interface without unnecessary steps.	Improves ease of use.

NFR- 6	Nonfunc.	The tool should be capable of handling files of different sizes without major performance issues.	Medium	The system successfully processes files of different supported sizes without major degradation in performance.	Supports varied input sizes.
NFR- 7	Nonfunc.	The implementation should be modular, with separate components for frequency analysis, tree construction, encoding, decoding, and file handling.	Medium	The implementation contains logically separated modules for the major compression and decompression operations.	Makes the system easier to maintain.
NFR- 8	Nonfunc.	The application should ideally be executable on commonly used systems without requiring major changes to the implementation.	Low	The application can be executed on supported commonly used systems with minimal or no implementation changes.	Improves portability.